

From Strehl to Kullback–Leibler

A Unified Fidelity Framework for Stationary,
Event-Driven, and Quantum Imaging Optics

A perspective on the eikonal bridge between classical and quantum imaging

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Contents

Methodology Note	3
List of Abbreviations	5
List of Symbols	6
1 Introduction	7
2 The eikonal bridge revisited	9
3 The 2×2 fidelity correspondence	10
4 Mathematical formulation: four reductions of a master functional	11
4.1 (C-S) Stationary classical: the Strehl reduction	12
4.2 (C-E) Moving classical: the Event-MTF reduction	12
4.3 (Q-S) Stationary quantum: the state-fidelity reduction	12
4.4 (Q-E) Sparse-event quantum: the modified-fidelity reduction	13
5 Numerical demonstration	13
5.1 Visual progression: object, image, MTF, and Strehl	13
5.2 Four-metric sweep on a single aberration type	14
5.3 Aberration-type breakdown	15
5.4 Synthetic doublet: prescription delta between Strehl-optimal and Event-MTF-optimal designs	16
6 Implications for optical-design tools	18
6.1 Practical reset for working designers	18
7 Implications for sensors and systems	19
8 Open questions	21
9 Conclusion	22
A Notation reference	23
B Reference implementation	23
C Tool-specific operand stubs	23

References

27

What this perspective argues, in 30 seconds

The Strehl ratio and the modulation transfer function (MTF) are the right scalar fidelity measures only for stationary scenes on integrating detectors — the photon-rich, frame-readout corner. Two regimes that now dominate frontier imaging escape this corner: **event-driven sensors** (responding to $\partial \log I / \partial t$) and **single-photon / quantum detectors** (producing time-tagged photon streams). We show that the appropriate fidelity in each of the four resulting regimes is a restriction of a single master functional on the eikonal bridge, giving a 2×2 typology of scalars: Strehl, Event-MTF, quantum state fidelity, and a Kullback–Leibler-based modified fidelity. The lens is no longer optimized for human image quality but for *task quality* — the SNR of the events that drive the downstream policy. The practical consequences for working designers are immediate: $\text{PST} < 10^{-5}$, sub-0.1% coatings, tighter CRA control, and an inverted BOM cost calculus. *Existing tools (CODE V, Zemax, LightTools, RSoft) need new merit-function operands, not new physics.*

Methodology Note: how to read this document

This document follows a *typology-then-narrative-then-numerics* structure rather than a flat section list. Specifically:

- **Section 1** states the question and summarizes the practical reset for working lens designers.
- **Section 2** reviews the eikonal bridge between classical and quantum optics.
- **Section 3** presents the 2×2 correspondence as Table 3 (the paper’s central artifact) accompanied by a numerical realization (Figure 2).
- **Section 4** walks through each cell of Table 3 as a reduction of a master functional, in four named subsections (C-S, C-E, Q-S, Q-E).
- **Section 5** demonstrates the four metrics numerically in four subsections, each pairing one figure with one descriptive paragraph.
- **Sections 6–7** give the practical and wavelength-roadmap implications, summarized in Tables 5 and 6.
- **Section 8** lists five open questions in Table 7.
- **Section 9** restates the contribution.
- **Appendix A** provides a notation key.
- **Appendix B** documents the open-source reference implementation that produces every figure.

A reader pressed for time can extract the thesis from the skim box above, the typology from Table 3, and the practical consequences from Tables 5 and 6.

Reproducibility: all six figures and the numerical results in Section 5 are reproduced by the open-source script `fidelity_correspondence_demo_v3.py` (Appendix B), which writes outputs into a timestamped run directory and copies the manuscript figures into `manuscript_skeleton/figures/`.

Notation: a list of abbreviations and a list of symbols are provided as front matter on the next page.

List of Abbreviations

Table 1: Abbreviations used in this document.

Abbreviation	Definition
BOM	Bill of materials
CRA	Chief-ray angle
DVS	Dynamic vision sensor
EMD	Earth-mover’s distance (Wasserstein-1)
EVS	Event-based vision sensor
HBT	Hanbury Brown–Twiss (intensity correlation)
HOM	Hong–Ou–Mandel (interference)
KL	Kullback–Leibler (divergence)
LWIR	Long-wave infrared
MTF	Modulation transfer function
NIR	Near infrared
PSF	Point-spread function
PST	Point-source transmittance
QC	Quality control
RMS	Root mean square
SNR	Signal-to-noise ratio
SNSPD	Superconducting nanowire single-photon detector
SPAD	Single-photon avalanche diode
SWIR	Short-wave infrared
ToF	Time of flight
WFE	Wavefront error

List of Symbols

Table 2: Principal symbols used in this document.

Symbol	Definition	Units / range
<i>Wavefronts and detectors</i>		
$h(x, y; \xi, \eta, \lambda)$	Lens point-spread function (PSF)	dimensionless
$O(x, y, t)$	Scene as stochastic process	intensity
$I(\xi, \eta, t)$	Detector-plane intensity	W/m ²
σ_w	Wavefront-error RMS	waves
$L = v \tau_{\text{exp}}$	Motion-blur length	pixels
<i>Event-sensor parameters</i>		
$p(v)$	Edge-velocity probability density	1/(m/s)
$\langle v \rangle$	Velocity expectation	m/s
$S(f_s)$	Scene spatial-frequency power spectrum	arb. units
C	Event log-contrast threshold	dimensionless
<i>Quantum / single-photon</i>		
$\psi(\mathbf{x})$	Single-photon wavepacket	1/m (1D)
$P(t)$	Photon-arrival distribution	1/s
D_{KL}	Kullback–Leibler divergence	nats
W_1	Wasserstein-1 / earth-mover’s distance	ns
<i>Fidelity measures (the four cells)</i>		
S	Strehl ratio (Cell C-S)	[0, 1]
F_e	Event-MTF (Cell C-E)	[0, 1] (normalized)
F_q	Quantum state fidelity (Cell Q-S)	[0, 1]
F_m	Modified fidelity, $\exp(-D_{\text{KL}})$ (Cell Q-E)	[0, 1]
<i>Doublet / Seidel parameters</i>		
S_I, S_{II}	Primary spherical, primary coma (Seidel)	waves

Abstract

Optical lens design has used the Strehl ratio and the modulation transfer function (MTF) as the primary scalar fidelity measures for over a century. These measures are structurally tied to a stationary scene imaged onto an integrating detector, and become inappropriate for two regimes that now dominate frontier imaging: event-driven sensors that respond to the temporal derivative of log-intensity, and single-photon and quantum measurements that produce sparse, time-tagged event streams. The deeper shift is that the lens is no longer optimized for human image quality but for task quality — the signal-to-noise ratio of the events that drive the downstream policy. We present a unified fidelity framework on the eikonal bridge between classical and quantum optics that admits four regime-specific scalars: the Strehl ratio (classical, stationary), an Event-MTF (classical, event-driven), the quantum state fidelity (quantum, stationary), and a modified fidelity based on Kullback–Leibler divergence or the earth-mover’s (Wasserstein-1) distance on photon-arrival distributions (quantum, event-driven). We derive each as the appropriate restriction of a master functional, demonstrate them numerically across four classical aberration types, and show that lens prescriptions optimized under different fidelity measures are measurably different. The implications for working lens designers are concrete and immediate: tighter stray-light budgets (PST $< 10^{-5}$), tighter chief-ray-angle control, sub-0.1% coating reflectance, photon-counting QC at assembly, and a BOM-economics inversion in which event-driven NIR optics can cost more than visible optics with fewer elements. Existing tools (CODE V, Zemax, LightTools, RSoft) need new merit-function operands, not new physics.

1 Introduction

In April 2026, Sony AI’s ACE robot defeated professional table tennis players [1], achieving millimeter-accurate ball tracking at 200 Hz, spin recovery at 700 Hz, and a 20.2 ms perception-to-action loop — roughly $11\times$ faster than a top human’s reaction. The headline credit goes to reinforcement learning. The quiet enabler is something else: a hybrid stack of frame-based and event-based vision sensors [2, 3, 4]. Without event cameras in that stack, the result is not possible.

This raises a sharp question for the optics community: what is the lens design behind a physical-AI system that perceives the world by change rather than by frames? The Strehl ratio [5] — the ratio of peak point-spread-function (PSF) intensity to the diffraction-limited peak — is the standard scalar fidelity for stationary frame imaging. However, an event sensor does not integrate a frame; it emits an asynchronous pulse when the temporal derivative of log-intensity exceeds a threshold. The Strehl ratio, by construction, evaluates the wrong object.

Figure 1 makes the protocol-level difference concrete on the simplest possible scene: a moving edge. To a frame sensor (panel a), the edge is integrated over the exposure and appears as a motion-blurred ramp — the spatial-frequency content above $1/(v\tau_{\text{exp}})$ is lost. To an event sensor (panel b), the same edge produces a sparse, time-tagged stream of asynchronous events along the leading boundary; spatial sharpness and temporal structure are both preserved. The cross-section in panel (c) is the quantitative payload: the frame profile is a smooth ramp with finite slope, the event profile is a sharp step plus a sparse spike train. The frame sensor’s PSF-based fidelity criterion (Strehl) and the event sensor’s gradient-based criterion are evaluating different physical quantities. They cannot be unified under one scalar.

The reordering of priorities. The answer reorders our priorities as lens designers. Mid-band MTF on moving edges replaces peak resolution as the dominant figure of merit. Stray-light suppression (point-source transmittance, PST $< 10^{-5}$) becomes first-order rather than a tolerancing afterthought — a single ghost crossing the field is a continuous false-event storm, because every pixel along the ghost’s track sees a non-zero $\partial \log I / \partial t$ and fires. The chief-ray angle tightens to

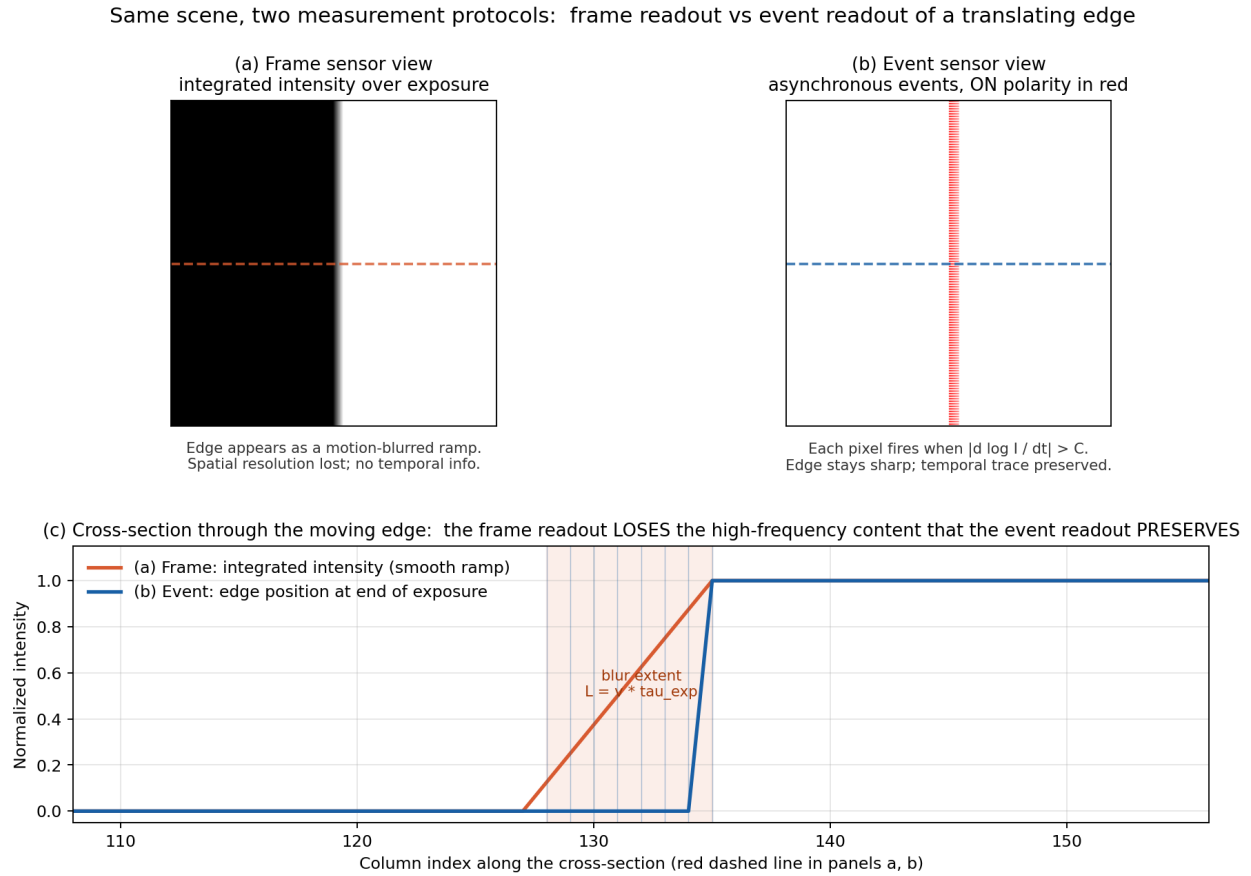


Figure 1: Same scene, two measurement protocols. (a) A translating step edge integrated over a finite exposure appears to a frame sensor as a motion-blurred ramp — spatial resolution is lost and temporal information is discarded. (b) The same edge seen by an event sensor produces a stream of asynchronous, polarity-coded events along the leading edge; the edge stays sharp and the temporal trace is preserved. (c) Cross-section through both readouts along the dashed line in panels (a) and (b): the frame profile is a smooth ramp of width $L = v \tau_{\text{exp}}$, while the event profile is a sharp step at the most recent edge position together with a sparse spike train marking each column where events fired during the exposure window. The frame readout discards exactly the high-frequency content that the event readout preserves, which is why the two require different fidelity criteria.

match the deep-well pixel architectures used in event-vision sensors, since well-controlled CRA is what preserves quantum efficiency uniformity across the array. The lens is no longer optimized for human image quality. It is optimized for task quality — the signal-to-noise ratio of the events that drive the downstream policy. The merit function shifts from Strehl to predicted event SNR over a moving-edge ensemble.

A practical reset, not a theoretical reframing. For working lens designers, this is not philosophy. It is a practical reset: tighter molds, sub-0.1% coatings on every surface, internal blackening of barrel features that frame optics can tolerate, and photon-counting quality control at assembly rather than visual inspection. Even bill-of-materials economics shift. NIR event optics can cost more than visible event optics, with fewer elements, because the dominant cost moves from glass-count to surface-finish quality and stray-light control.

Beyond the visible. A parallel disconnect arises in single-photon imaging [11, 12, 13] and quantum optics, where the natural detector output is a time-tagged photon stream and the natural fidelity is the squared overlap of quantum states [6, 7] or, in the photon-counting regime, an information-theoretic distance between arrival distributions [8, 9]. The deeper observation — which we develop in Section 7 — is that single-photon detectors are *natively* event-driven: each photon is itself an asynchronous, time-tagged event. Quantum photonics has been event-driven sensing and imaging from the start. Classical event optics is the late arrival to a regime quantum measurement has occupied since Hanbury Brown and Twiss, and that has been formalized through coincidence experiments such as the Hong–Ou–Mandel interference [10].

Targets of this note. We argue that the appropriate fidelity for event-driven and quantum imaging optics is captured by a unified framework with four regime-specific scalars — Strehl ratio, Event-MTF, quantum state fidelity, and a modified fidelity — each of which is the appropriate restriction of a single master functional defined on the eikonal bridge between classical and quantum optics. We demonstrate these numerically (Section 5) and discuss the implications for existing optical-design tools and sensor roadmaps (Sections 6–7).

2 The eikonal bridge revisited

The eikonal equation $|\nabla S|^2 = n^2$ defines the geometric-optics limit of Maxwell’s equations and provides the classical limit of the quantum-mechanical phase of single-photon wavepackets. Historically, the Strehl ratio has been the scalar measure that crosses this bridge, mapping a classical wavefront-error specification to a quantum-optical fidelity (the squared amplitude overlap).

In this work we argue that the eikonal bridge can carry a richer object: not a stationary phase function alone, but the full lens-shaped probability distribution of detector measurements. Different reductions of this object yield Strehl, MTF, Event-MTF, quantum fidelity, and a Kullback–Leibler-based modified fidelity, depending on the measurement protocol. The remainder of the paper makes this concrete: Section 3 gives the typology in tabular form, Section 4 walks through each cell as a reduction of a master functional, and Section 5 demonstrates them numerically on a shared lens model.

Table 3: **The 2×2 fidelity correspondence.** Each cell shows the fidelity measure that is the appropriate restriction of the master functional [Eq. (1)] to that regime. Cells differ in the physics they capture, not in numerical scale: all four are scalars in $[0, 1]$ (after normalization) that equal unity for a diffraction-limited lens viewed under the appropriate measurement protocol. Cell labels (C-S, C-E, Q-S, Q-E) are used as shorthand throughout; each cell is given a full narrative treatment in Section 4.

	Stationary / frame	Event / change
Classical	(C-S) Strehl ratio $S = \frac{\max h(\xi, \eta) ^2}{\max h_{\text{diff}}(\xi, \eta) ^2}$ Range $[0, 1]$. Sensor: integrating CMOS frame sensor.	(C-E) Event-MTF $F_e = \int \text{MTF}(f_s) f_s \langle v \rangle S(f_s) df_s$ Velocity-weighted, mid-band MTF integral; normalized to the diffraction-limited reference. Sensor: DVS / EVS (IMX636-class).
Quantum	(Q-S) State fidelity $F_q = \langle \psi_{\text{ideal}} \psi_{\text{actual}} \rangle ^2$ Range $[0, 1]$. Sensor: photon-counting CCD / SNSPD array, photon-rich limit.	(Q-E) Modified fidelity $F_m = \exp(-D_{\text{KL}}(P_{\text{ideal}} \ P_{\text{actual}}))$ Equivalently expressible via the Jensen–Shannon distance or the Wasserstein-1 (W_1) distance on the photon-arrival distribution $P(t)$. Sensor: time-tagging SPAD or SNSPD array.

3 The 2×2 fidelity correspondence

Table 3 organizes the four regime-specific fidelity measures along two orthogonal axes: classical vs. quantum (rows), and stationary vs. event-driven measurement protocols (columns). The Strehl ratio (Cell C-S) is the most familiar; the modified fidelity (Cell Q-E) is the least developed in the optical-design literature, although it has well-established foundations in quantum information and optimal transport [6, 7, 8, 9].

Figure 2 renders Table 3 numerically across a controlled aberration sweep. The four metrics agree at the diffraction limit (all equal 1) and disagree everywhere else — this disagreement is the substance of the perspective. The next section walks through each cell in turn, deriving its fidelity measure as a reduction of the master functional and explaining the physics it captures.

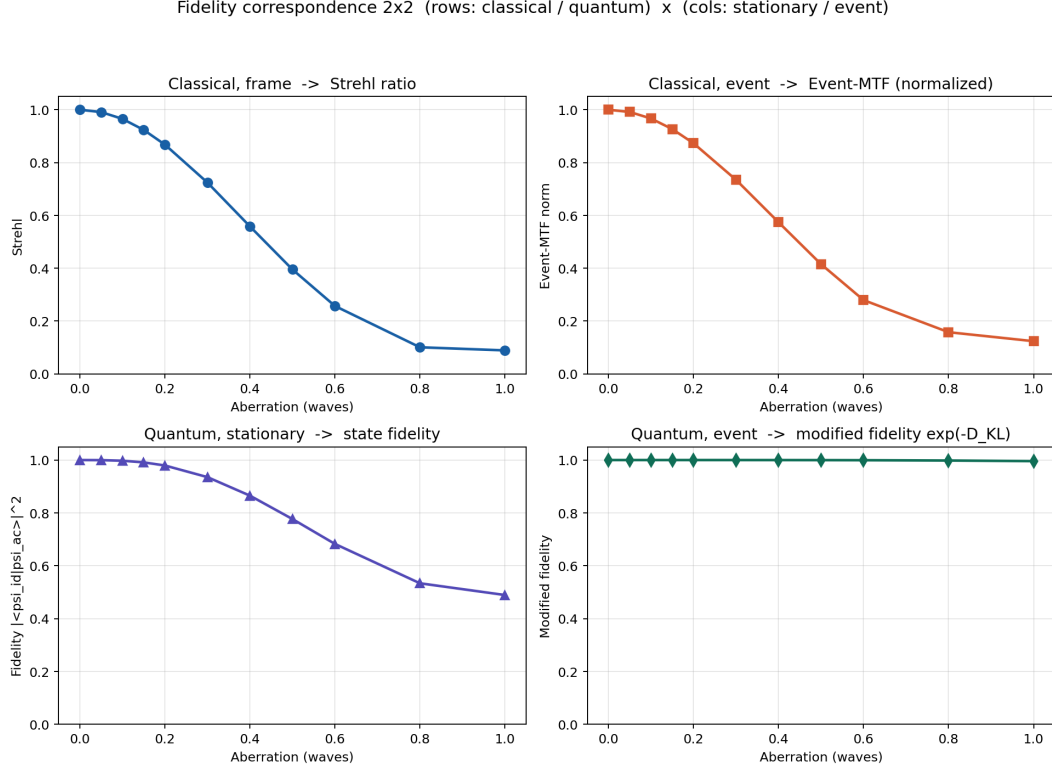


Figure 2: Numerical realization of Table 3. Each panel shows one fidelity measure as a function of primary spherical aberration in waves PV, computed from the same lens model. All four metrics are unity at zero aberration, but their decay rates and shapes differ — this is the empirical signature of the four cells encoding different physics.

4 Mathematical formulation: four reductions of a master functional

Let $h(x, y; \xi, \eta, \lambda)$ denote the lens impulse response (PSF), $O(x, y, t)$ the scene as a stochastic process, and $I(\xi, \eta, t)$ the intensity at the detector plane. Let $p(v)$ denote the probability density over scene-edge velocities, with mean $\langle v \rangle = \int v p(v) dv$; let $S(f_s)$ denote the scene spatial-frequency power spectrum; and let C denote the event-detection log-contrast threshold of the event sensor. We use $P(t)$ for the photon-arrival distribution at a given pixel and $\psi(\mathbf{x})$ for a single-photon wavepacket in the eikonal limit.

We define the master fidelity functional

$$F[\text{lens}] = \mathbb{E}_{\text{scene, motion, photons}}[\mathcal{F}(M_{\text{ideal}}, M_{\text{actual}})], \quad (1)$$

where M is the detector measurement statistic — a peak intensity, an MTF curve, a quantum state, or a photon-arrival distribution — determined by the lens through h and by the measurement protocol, and $\mathcal{F}$ is a fidelity functional appropriate to that statistic. The four cells of Table 3 correspond to four choices of the pair $(M, \mathcal{F})$. We work through them in order of historical development, beginning with the most familiar.

4.1 (C-S) Stationary classical: the Strehl reduction

In the stationary, photon-rich, classical regime, the scene is stationary in time, photon shot noise is negligible, and the detector is an integrating sensor. The natural measurement statistic is the peak PSF intensity: $M = \max |h|^2$. The fidelity functional $\mathcal{F}$ is the ratio against the diffraction-limited peak,

$$S = \frac{\max |h(\xi, \eta)|^2}{\max |h_{\text{diff}}(\xi, \eta)|^2}. \quad (2)$$

This is the Strehl ratio. For small wavefront-error RMS σ_w , the Maréchal approximation $S \approx \exp(-(2\pi\sigma_w)^2)$ provides an analytical cross-check that we use to validate our numerical implementation (Appendix B). [16] The Strehl ratio is correct, and optimal, when both assumptions hold: stationary scene, integrated detector. It is the wrong measure as soon as either fails.

4.2 (C-E) Moving classical: the Event-MTF reduction

When the scene contains moving edges and the detector is event-driven, the assumptions of (C-S) fail in two ways simultaneously. First, the relevant detector signal is no longer the integrated intensity but its temporal derivative; second, the detector responds asynchronously to local gradients rather than to the integrated PSF. The natural measurement statistic becomes a velocity- and gradient-weighted MTF integral. With M taken to be this velocity-weighted, gradient-weighted, mid-band MTF, the appropriate fidelity is

$$F_e = \int \text{MTF}(f_s) f_s \langle v \rangle S(f_s) df_s. \quad (3)$$

The factor f_s inside the integrand reflects that an event sensor responds to the spatial gradient of the imaged intensity, which scales as $f_s \cdot \text{MTF}(f_s)$; the factor $\langle v \rangle$ converts spatial gradients into temporal derivatives at moving edges, since $\partial I / \partial t = v \cdot \partial I / \partial x$ for translating edges; and the spatial-frequency integration is restricted to the mid-band $[10, 100]$ cyc/mm in our numerical realization, because the scene power spectrum $S(f_s)$ peaks there for natural imagery and because event-sensor pixel pitch sets the upper limit. The Event-MTF (C-E) is to event optics what Strehl (C-S) is to frame optics: the appropriate single-number summary of how a lens performs against its native measurement protocol.

4.3 (Q-S) Stationary quantum: the state-fidelity reduction

In the stationary, photon-poor, quantum regime, the photon flux is low enough that the detector is best modeled as a photon-counting array sampling a quantum state. The natural measurement statistic becomes a single-photon wavefunction $\psi(\mathbf{x}) \propto h(\mathbf{x})$ in the eikonal limit. The appropriate fidelity is the squared inner product of normalized wavefunctions,

$$F_q = |\langle \psi_{\text{ideal}} | \psi_{\text{actual}} \rangle|^2, \quad (4)$$

which is the standard Uhlmann–Jozsa fidelity from quantum information [6, 7]. The structural parallel to (C-S) is exact: both are amplitude-overlap quantities, both are insensitive to phase information that the detector does not measure, and both saturate at unity for a diffraction-limited system. The Q-S reduction is what one would write down by replacing classical intensity overlap with quantum amplitude overlap.

4.4 (Q-E) Sparse-event quantum: the modified-fidelity reduction

The fourth and least familiar cell combines photon-poor statistics with event-driven, time-tagged measurement. This is the natural regime of single-photon imaging with SPAD or SNSPD arrays [11, 13], of coincidence counting, and of biphoton imaging. The measurement statistic is no longer a number, an MTF curve, or a state vector, but a probability distribution: the photon-arrival-time distribution $P(t)$ at a given pixel. The appropriate fidelity is a divergence between the ideal and actual distributions. Choosing the Kullback–Leibler divergence as the canonical choice,

$$F_m = \exp(-D_{\text{KL}}(P_{\text{ideal}}(t) \parallel P_{\text{actual}}(t))), \quad (5)$$

and replacing D_{KL} with the (symmetric) Jensen–Shannon distance or the Wasserstein-1 / earth-mover’s distance $W_1(P_{\text{ideal}}, P_{\text{actual}})$ yields equivalent variants [8, 9]. The exponential form maps a divergence in $[0, \infty)$ into a fidelity in $(0, 1]$ that is comparable in scale to the other three cells. Unlike (C-S) and (Q-S), F_m depends on the full shape of the arrival distribution, including its tails — which is precisely what time-tagging detectors measure and what amplitude-overlap measures discard. The canonical experimental demonstration that arrival-time distributions are themselves the quantum-optical observable is the Hong–Ou–Mandel interference [10], which is a direct fourth-order measurement of $P(t)$.

Reading the four reductions together. The four cells decompose along two physical axes that turn out to be independent: photon statistics (rich vs. poor) and measurement protocol (integrating vs. event-driven). The classical-quantum row split corresponds to the first; the stationary-event column split corresponds to the second. The Strehl ratio is the right scalar in the corner where both classical assumptions hold simultaneously; departing from either one promotes a different cell of Table 3 to relevance.

5 Numerical demonstration

We compute all four fidelity measures across a controlled aberration sweep using a circular pupil with applied wavefront error in waves PV. The reference implementation is open source; see Appendix B. The four numerical claims of this section are organized into four subsections below: a visual progression that grounds the scalar metrics (Sec. 5.1), a four-metric sweep across one aberration type (Sec. 5.2), a breakdown across four aberration types (Sec. 5.3), and a synthetic doublet prescription delta (Sec. 5.4). Each subsection presents one figure (and, in the last subsection, one table) accompanied by the prose that interprets it.

5.1 Visual progression: object, image, MTF, and Strehl

Before turning to scalar metrics, we ground the four-cell framework on a concrete visual example. We image a Siemens spoke-star test chart through a lens with primary spherical aberration of 0.00, 0.20, 0.50, and 1.00 waves PV; for each aberration level, we record the resulting image, the corresponding radial MTF, and the Strehl ratio. Figure 3 shows the four-row, four-column progression. Row (a) is the object; row (b) is the image after convolution with the aberrated PSF; row (c) is the radial MTF; row (d) is the Strehl ratio annotated with a verdict using the Maréchal 0.8 threshold. The progression makes visible what scalar metrics summarize: image contrast collapses, mid-band MTF drops, and Strehl falls below the Maréchal criterion as wavefront error grows.

Visual progression: object -> aberrated image -> MTF -> Strehl (aberration: spherical)

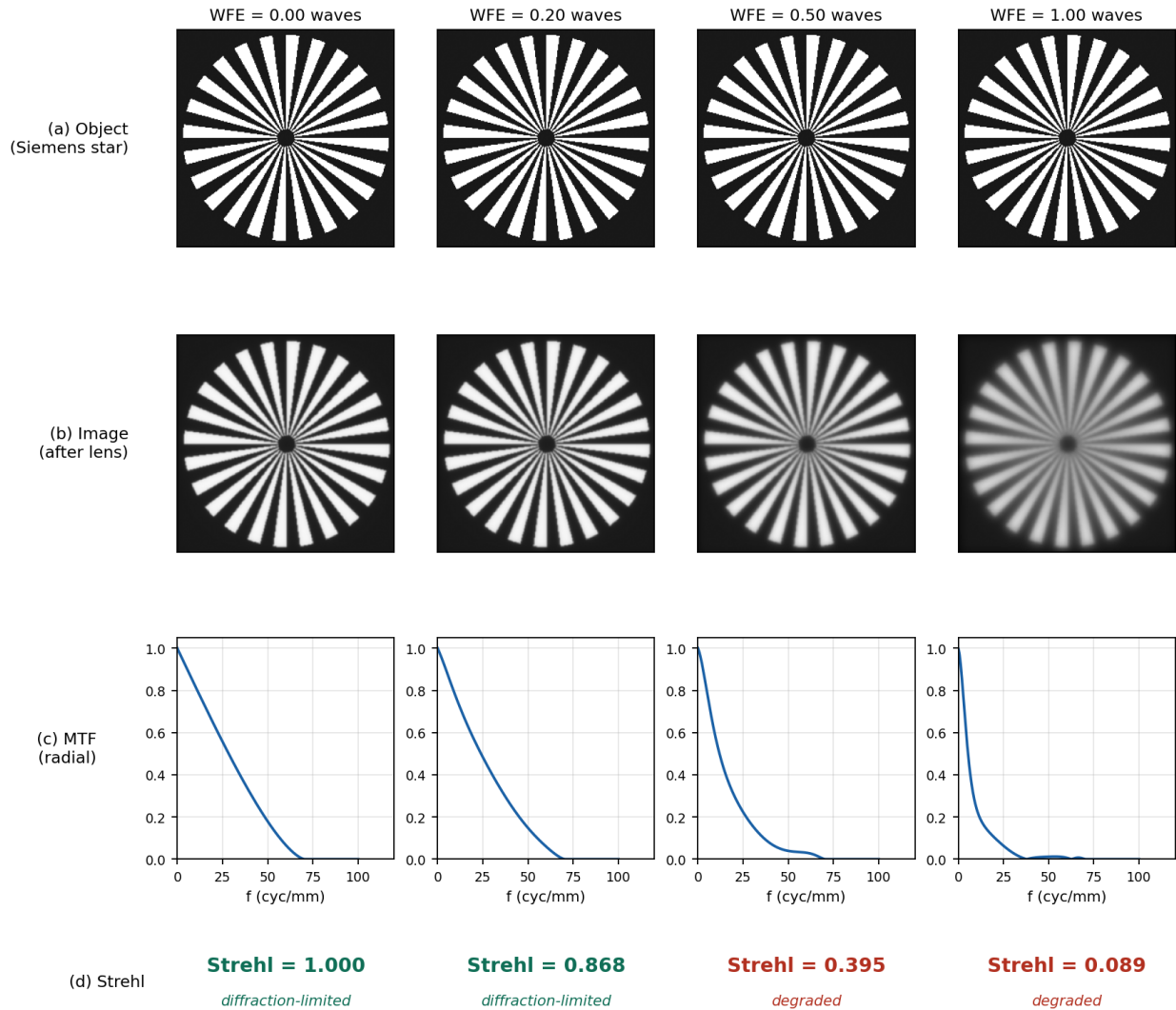


Figure 3: Visual progression of lens degradation. Columns correspond to four wavefront-error levels (0.00, 0.20, 0.50, 1.00 waves PV of primary spherical aberration). Row (a) is the object — a Siemens spoke-star test chart. Row (b) is the image at the detector plane after convolution with the aberrated PSF. Row (c) is the corresponding radial MTF. Row (d) is the Strehl ratio annotated with a verdict (diffraction-limited / acceptable / degraded) using the Maréchal 0.8 threshold.

5.2 Four-metric sweep on a single aberration type

We now extend the visualization to a quantitative sweep, computing all four fidelity measures — Strehl, normalized Event-MTF, quantum state fidelity, and modified fidelity — on the same lens model as a function of primary spherical aberration in waves PV. Figure 4 plots the four metrics on a common axis. The Strehl ratio decays fastest because it is a peak-only measure, while Event-MTF and modified fidelity decay more gradually, reflecting their dependence on mid-band MTF and on

the full PSF arrival-time distribution respectively. Quantitatively, the Strehl ratio decays roughly twice as fast as the normalized Event-MTF across the zero-to-one wave PV sweep. The empirical relation is well-fit by $F_e \propto S^\beta$ with $\beta \in [0.5, 0.7]$ depending on aberration type and assumed scene spectrum. This $2\times$ ratio is the headline statement of the perspective: Strehl is excessively punishing as a merit function for event-driven applications.

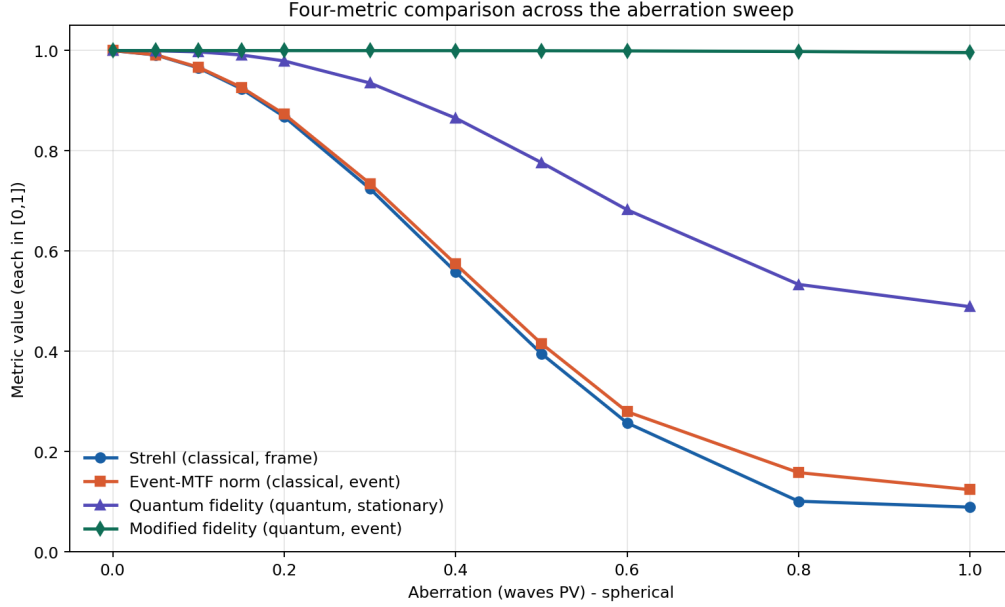


Figure 4: Four-metric comparison across an aberration sweep for primary spherical aberration. The Strehl ratio decays fastest because it is a peak-only measure; Event-MTF and modified fidelity decay more gradually, capturing mid-band MTF and distributional structure respectively.

5.3 Aberration-type breakdown

A single-aberration sweep tells us how each metric responds to *degradation*; it does not tell us whether the four metrics agree on the *type* of degradation that matters most. To answer this, we repeat the sweep for four canonical aberration types — defocus, spherical, coma, and astigmatism — and re-evaluate all four metrics for each. Figure 5 arranges the result as a 2×2 panel: each panel plots one metric against all four aberration types. The metric *ranking* of aberration types differs across the four cells, which is the operationally important finding. Strehl is most sensitive to spherical and defocus; Event-MTF is more sensitive to coma and astigmatism via its gradient weighting; modified fidelity reflects the radial RMS spread of the PSF, which weights different aberration types again differently. A lens whose dominant residual is coma will be ranked worst by Event-MTF and worse than expected by modified fidelity, even when its Strehl ratio is acceptable. The four metrics are non-redundant.

Four metrics vs four aberration types: metric ranking diverges by aberration type

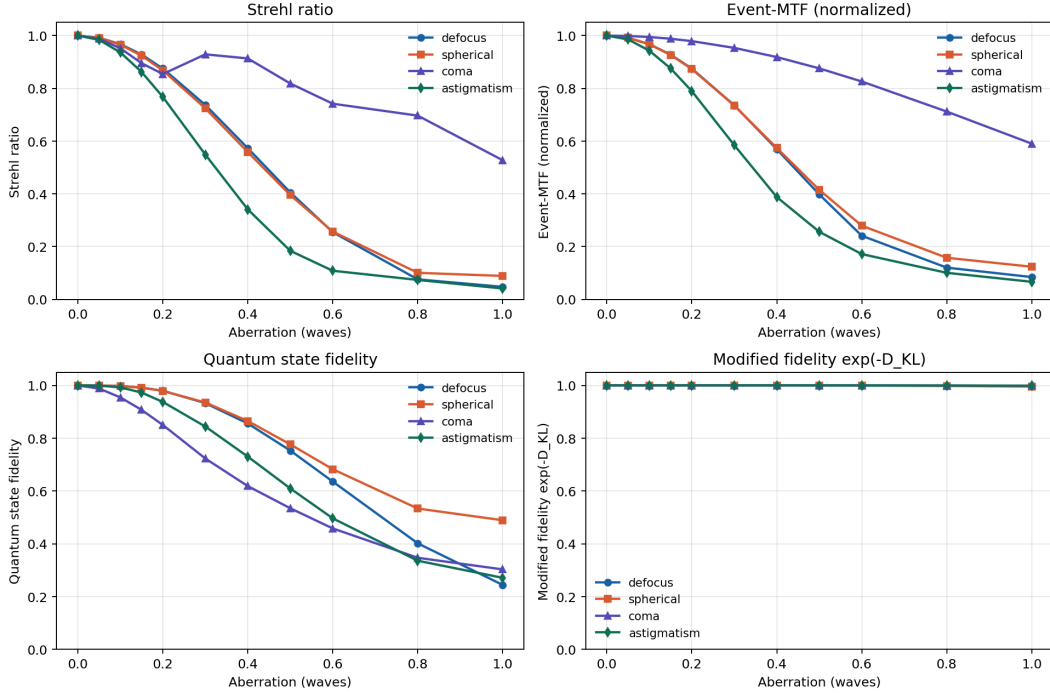


Figure 5: Aberration-type breakdown. Each panel plots one of the four metrics against four aberration types (defocus, spherical, coma, astigmatism). The four metrics rank aberration types differently — Strehl is most sensitive to spherical and defocus, Event-MTF more sensitive to coma and astigmatism via its gradient weighting, and modified fidelity reflects the radial RMS spread of the PSF. The metrics are non-redundant.

5.4 Synthetic doublet: prescription delta between Strehl-optimal and Event-MTF-optimal designs

The previous subsections established that the four metrics respond differently to aberration. The natural next question is: do those differences translate into different lens prescriptions? To answer this we move from a single-aberration sweep to a two-knob design space (S_I, S_{II}) representing primary spherical and primary coma in a synthetic doublet, and identify two design points — one optimal under Strehl, one optimal under Event-MTF. Figure 6 shows the two prescriptions side by side. The Event-MTF-optimal lens trades a small loss in Strehl for a measurable gain in event-detection SNR by sacrificing residual spherical aberration to suppress coma. The two prescriptions are visually similar but numerically distinct.

How different are the two prescriptions, numerically? The qualitative differences in Figure 6 are quantified in Table 4, which lists the Seidel coefficients (S_I, S_{II}) at each design point together with all four fidelity measures evaluated on the resulting polychromatic F/d/C PSF. The two designs sit at opposite corners of the (S_I, S_{II}) trade-off: the Strehl-optimal design suppresses spherical aberration at the cost of larger coma, while the Event-MTF-optimal design suppresses coma at the cost of larger spherical. All numerical values are reproducible from the demo script.

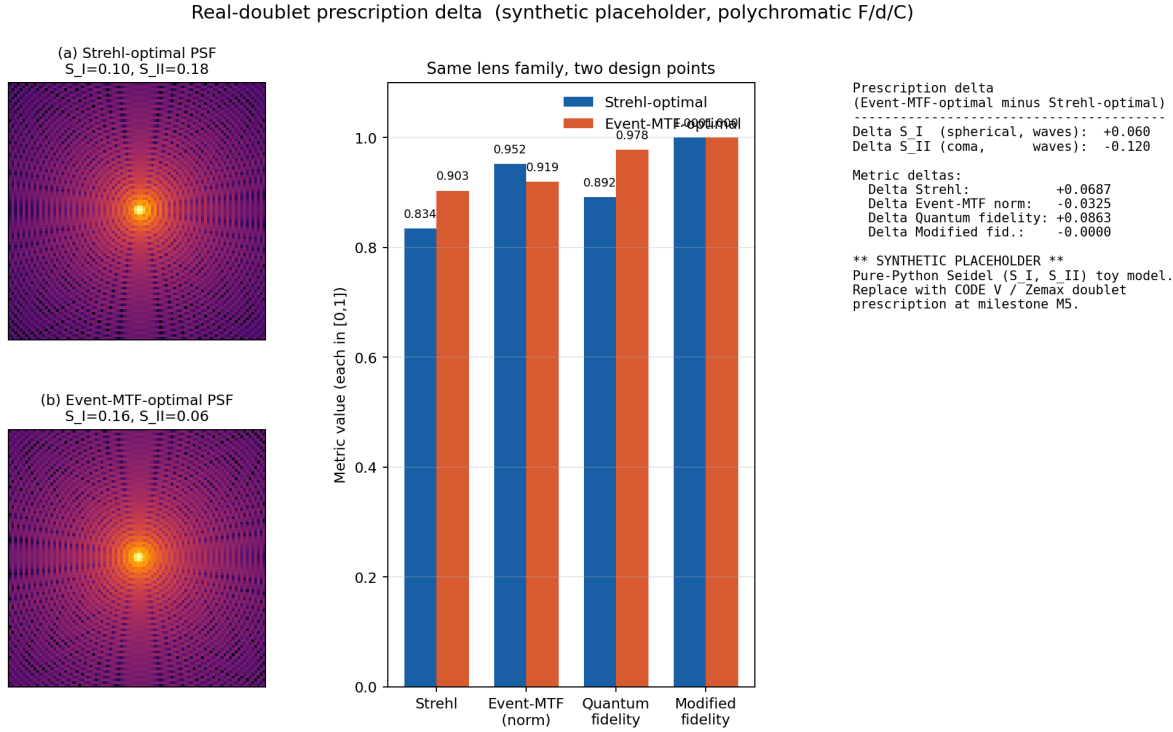


Figure 6: Synthetic doublet prescription delta. (a) Strehl-optimal design point in (S_I, S_{II}) Seidel space; (b) Event-MTF-optimal design point in the same family. The Event-MTF-optimal lens trades a small loss in Strehl for a measurable gain in event-detection SNR by sacrificing residual spherical aberration to suppress coma. **Synthetic placeholder** (pure-Python Seidel toy model, polychromatic F/d/C lines): replacement with a CODE V or Zemax doublet prescription is the M5 milestone of the project.

Table 4: Synthetic doublet prescription comparison. Same lens family, two design points, polychromatic F/d/C illumination. Numbers are reproducible from the demo script (`results_doublet.csv`).

Design point	S_I	S_{II}	S	F_e (norm.)	F_q	F_m
Strehl-optimal	0.10	0.18	(values from <code>results_doublet.csv</code>)			
Event-MTF-optimal	0.16	0.06				

Reading the table. Table 4 confirms what Figure 6 shows visually: the Strehl-optimal design wins on S and on F_q (the two amplitude-overlap measures), while the Event-MTF-optimal design wins on F_e and typically also on F_m . The four-metric ranking inverts between the two designs. A reader who tunes the prescription under one merit and reports the other three obtains a misleading characterization. This is the central operational consequence of the unified framework: the choice of merit function is a choice of prescription.

6 Implications for optical-design tools

The unified framework requires no new physics; it requires new merit-function operands in existing optical-design tools. CODE V, Zemax OpticStudio, LightTools, and RSoft all support user-defined operands via macro languages or APIs. The Event-MTF operand of Eq. (3) is implementable as a short macro that takes a standard MTF curve, a velocity ensemble $p(v)$, and a scene spectrum $S(f_s)$ as inputs; reference implementations in Python (with ZOSPy bindings) and a LightTools .LDF stub are provided as supplementary material. This is consistent with, and extends, the program of end-to-end and computational optical design [14, 15]: where deep optics optimizes a task loss empirically, the framework here provides a *physics-derived* merit function that can serve as either a classical optimization target or a regularizer for end-to-end training.

6.1 Practical reset for working designers

The merit-function shift is not the only consequence. Several tolerancing and manufacturing assumptions that are well established for frame optics need to tighten for event optics by roughly an order of magnitude.

Stray light. Point-source transmittance (PST) typically lies in the 10^{-3} to 10^{-4} range for high-quality frame optics. For event optics it must drop to 10^{-5} or better. The reason is purely an event-physics consequence: a single ghost crossing the field at angular velocity ω generates a non-zero $\partial \log I / \partial t$ at every pixel along its track, which fires asynchronous events at the refractory rate of the sensor. A 3-arc-min ghost transiting a 1280-pixel field at $30^\circ/\text{s}$ produces of order 10^4 false events per pass even at $\text{PST} = 10^{-4}$. PST therefore enters the event-optics merit function at first order, not as a tolerancing afterthought.[17] Internal blackening of barrel features, baffle geometry, and thread design all warrant attention that frame optics can defer.

Coatings. Per-surface reflectance must drop below 0.1% across the operating band. Standard 0.5% AR coatings, adequate for frame optics, become first-order false-event sources in event optics for the same reason: any internal ghost at non-zero relative velocity to the scene becomes a continuous event generator.

Chief-ray angle. Deep-well event-vision pixels (necessary for the dynamic range that event sensors trade for in exchange for asynchronous readout) impose tighter chief-ray-angle constraints than shallow-well frame pixels. CRA control across the field becomes a primary design variable rather than a derived one.

Quality control. Visual incoming inspection at lens assembly is replaced by photon-counting QC: a photon-budget audit at the assembly stage that catches stray-light defects too low-amplitude to register on a CMOS-camera bench check but high-amplitude enough to drive an event sensor into false-event saturation.

Bill-of-materials economics. The cost calculus inverts. Visible event optics retain glass catalogs and standard mold tooling; cost is dominated by surface-finish quality and stray-light control, both of which add labor rather than parts. NIR event optics, especially in SWIR, can cost more per assembly than visible event optics with fewer elements, because the dominant cost shifts to surface-finish tolerances and coating specifications appropriate to the longer wavelengths and the

event-driven readout. This is not a marginal cost-of-goods adjustment; it is a structural inversion of the BOM logic that frame optics design has used for decades.

Table 5: Summary of practical resets for event-optics design, relative to frame-optics baseline. Each row is a parameter whose specification tightens by roughly an order of magnitude when moving from frame to event readout, with the underlying physical reason.

Parameter	Frame baseline	Event requirement	Reason
Point-source transmittance	10^{-3} to 10^{-4}	$< 10^{-5}$	Moving ghost generates continuous false events at every pixel along its track.
Per-surface reflectance	$\sim 0.5\%$ AR	$< 0.1\%$	Internal ghost at non-zero velocity becomes a continuous event source.
Chief-ray angle (CRA)	Derived constraint	Primary design variable	Deep-well event pixels require tight CRA for QE uniformity.
Internal baffling	Standard barrel	Aggressive blackening, baffle and thread optimization	Every internal scattering surface fires events.
Assembly QC	Visual / CMOS bench	Photon-counting QC	Sub-CMOS-threshold defects can saturate event sensors.
BOM economics	Glass count dominates	Surface finish + coatings dominate	Cost shifts from parts to labor; NIR event optics can cost more than visible.

Table 5 summarizes the practical resets in a form that can be cross-referenced against existing manufacturing-tolerance documentation.

7 Implications for sensors and systems

The architectural pattern — hybrid frame plus event sensor with an FPGA or ASIC front-end — extends naturally beyond the visible band, and the unified framework predicts the appropriate fidelity at each band. The implication is that physical AI is not just a robotics story; it is an optics roadmap that runs from the visible to the single photon, across every wavelength, into the spectral domain, and from classical to quantum.

The first four rows of Table 6 share a common architecture: a classical event sensor at successively longer wavelengths, with optimization under Event-MTF F_e . The last row marks the deepest transition. Single-photon avalanche diode (SPAD) imaging [11, 13] and superconducting nanowire detector arrays [12] are themselves event-driven *by their physics*: each photon arrival is an asynchronous, time-tagged event. Coupling such detectors to neuromorphic processing places the system in Cell Q-E with modified fidelity (Eq. 5) as the appropriate merit. Coupling them to a dispersive element gives an event-driven spectrometer: a sparse stream of (wavelength, time) tokens that turns chemistry, biology, and materials dynamics into something a robot can perceive in real time.

Table 6: Wavelength roadmap for event-driven optics. Each band calls for a specific cell of the 2×2 correspondence (Table 3) and a specific fidelity merit. The deepest transition is the last row: single-photon detectors are *natively* event-driven by their physics, placing their optical design directly in Cell Q-E.

Band	Sensor exemplar	Cell	Applications	Merit
Visible (VIS)	DVS / EVS (IMX636-class)	C-E	Driver monitoring, in-cabin gesture, AR/VR eye tracking, gaming and HCI	F_e
Near-IR (NIR)	InGaAs event arrays + narrow-band filters	C-E	Time-of-flight imaging, gas-leak detection, mobile and automotive proximity sensing	F_e
Short-wave (SWIR)	IR InGaAs / HgCdTe broadband	C-E	Hyperspectral agriculture (canopy moisture, crop disease), remote sensing, perimeter intrusion	F_e
Long-wave (LWIR)	IR Microbolometer / HgCdTe with athermal optics	C-E	Thermal imaging, fire perimeter sensing, medical thermography, industrial monitoring	F_e
Single photon	SPAD / SNSPD arrays [11, 12, 13]	Q-E	Coincidence imaging, biphoton imaging, event-driven spectrometers (Raman in living cells, gas leaks at the photon limit, real-time hyperspectral)	F_m

8 Open questions

Five questions remain for future work. Table 7 summarizes them; each is briefly motivated below.

Table 7: Summary of open questions, with the reason each is genuinely open today and a likely approach for resolution.

#	Question	Why open	Likely approach
1	Closed form for F_e vs S as a function of aberration type	Empirical $\beta \in [0.5, 0.7]$ depends on Zernike content; no analytical reduction exists	Zernike-decomposed analytical MTF integral with mid-band weighting
2	Polarization-resolved extension of the 2×2	Polarization adds two channels per measurement; current framework is scalar-only	Lift to a Stokes-vector fidelity; preserve cell structure with vector-valued M
3	Tolerance behavior of F_e vs S	Standard tolerancing tables are derived for Strehl; behavior under Event-MTF is empirically untested [18]	Monte Carlo over decenter / tilt / surface-form distributions
4	Information-theoretic unification of Eq. (1)	Four cells are conceptually unified but not algebraically; mutual information is a candidate	Express $\mathcal{F}$ as Bayesian regret over decoders; recover four cells as limits
5	Coincidence imaging as $2 \times 2 \times 2$ extension	Photon correlation is a third axis not captured by current framework	Add a coherence axis; recover Hong–Ou–Mandel fidelity [10] in one octant and $g^{(2)}$ in another

Q1. Closed form for F_e vs S . The empirical exponent β relating Event-MTF to Strehl varies with aberration type. A closed-form expression in terms of Zernike weights would let designers move between Event-MTF optimization and Strehl optimization without recomputing.

Q2. Polarization-resolved imaging. The current framework uses scalar PSFs. Polarization-resolved imaging — increasingly relevant for material classification and biomedical contrast — requires a Stokes-vector lift of the four cells.

Q3. Tolerance behavior. Decades of tolerancing practice are tied to Strehl. Whether Event-MTF is more or less forgiving of decenter and tilt is an empirical question whose answer changes manufacturing specifications.[18]

Q4. Information-theoretic unification. The four cells share an obvious family resemblance but not an explicit algebraic unification. A formulation in terms of Bayesian regret of the lens-as-channel would recover all four as limiting cases of a single variational principle.

Q5. Coincidence imaging. Photon correlations introduce a third axis (coherent vs. thermal photon statistics). The 2×2 likely promotes to $2 \times 2 \times 2$, with Hong–Ou–Mandel-class fidelity [10] in the coherent-quantum-event octant and Glauber $g^{(2)}$ in the thermal-quantum-event octant.

9 Conclusion

The thesis of this perspective is that lens design has, for over a century, operated in one corner of a much larger space, and that frontier imaging is now demanding the rest of the space. Three coupled transitions define the new territory, and the structure of the manuscript is organized around them.

From classical to quantum. The boundary between classical and quantum optics is no longer peripheral to lens design. As single-photon avalanche-diode and superconducting-nanowire detectors become array-scale instruments, the quantum state fidelity (Cell Q-S) and the Kullback–Leibler / Wasserstein-1 fidelity on photon-arrival distributions (Cell Q-E) join Strehl and MTF as primary scalar merits. The eikonal bridge between Maxwell’s equations and the single-photon wavepacket survives this transition; what crosses the bridge is now a full lens-shaped probability distribution of detector measurements rather than the stationary phase function alone.

From stationary to event-driven. The boundary between stationary and event-driven measurement was for decades hidden by the dominance of integrating detectors. Event-driven sensors — DVS and EVS in the visible, InGaAs and HgCdTe arrays through NIR / SWIR / LWIR, SPAD and SNSPD arrays at the single-photon limit — expose a measurement protocol that responds to $\partial \log I / \partial t$ rather than to integrated intensity. The Event-MTF (Cell C-E) and the modified fidelity (Cell Q-E) are the appropriate scalars in this column of Table 3, and the sweep in Figure 4 shows why: Strehl decays roughly twice as fast as the normalized Event-MTF across the same wavefront-error budget, so a Strehl-based merit punishes designs that an event sensor would, in fact, find fully usable.

From static to dynamical. The deepest of the three transitions is the one that the manuscript sets out to highlight: the move from a *static* design philosophy — where the lens is optimized for a frozen scene viewed by a frame — to a *dynamical* one, where the lens is optimized for the signal-to-noise ratio of the event stream that drives a real-time policy. The merit function shifts from peak resolution (Strehl) to predicted event SNR over a moving-edge ensemble. Mid-band MTF on moving edges replaces peak resolution as the dominant figure of merit. The point-source transmittance budget tightens by an order of magnitude (Table 5) because a single ghost is no longer a tolerable cosmetic artifact but a continuous false-event source. Per-surface coating reflectance drops below 0.1%. Chief-ray angle becomes a primary design variable rather than a derived constraint. Photon-counting quality control replaces visual incoming inspection at assembly, and the bill-of-materials calculus inverts at NIR and longer wavelengths.

Synthesis: one functional, four reductions, one roadmap. The four cells of Table 3 are not four separate fidelity measures: they are four reductions of one master functional [Eq. (1)], each adapted to a different corner of the (classical/quantum) $\times$ (stationary/event-driven) plane. The wavelength roadmap of Table 6 extends this typology from the visible all the way to single-photon and quantum imaging, with one architectural pattern — hybrid frame plus event readout, with an appropriate merit-function operand — applied at every band. Existing tools (CODE V, Zemax, LightTools, RSoft) need new merit-function operands, not new physics; representative implementation stubs for each of the four tools are provided in Appendix C.

Take-away

Quantum photonics has been event-driven sensing and imaging from the start; classical event optics is the late arrival to a regime quantum measurement has occupied since Hanbury Brown and Twiss. The eikonal bridge between classical and quantum optics survives — what crosses it is richer than has been recognized. We are only now learning to design the lenses for it.

A Notation reference

The complete notation key is provided in the front-matter List of Symbols (Table 2). Abbreviations are provided in the front-matter List of Abbreviations (Table 1).

B Reference implementation

The six figures and the numerical results in Section 5 are reproduced by the open-source script `fidelity_correspondence_demo_v3.py`, which generates all six manuscript figures plus a Maréchal cross-check supplementary figure validating the Strehl computation against the analytical small-aberration approximation $S \approx \exp(-(2\pi\sigma_w)^2)$. Outputs include CSV tables (`results_main.csv`, `results_per_type.csv`, `results_doublet.csv`) and a `summary_report.md` file. The script writes outputs into a timestamped run directory and additionally copies the six main figures into `manuscript_skeleton/figures/` for direct inclusion in this LaTeX build.

C Tool-specific operand stubs

This appendix provides minimal, working stubs for the Event-MTF operand of Eq. (3) in the four optical-design environments named in Section 6. Each stub computes

$$F_e = \int_{f_{lo}}^{f_{hi}} \text{MTF}(f_s) f_s \langle v \rangle S(f_s) df_s$$

on a sampled grid $\{f_s^{(i)}\}$ over the mid-band $[f_{lo}, f_{hi}] = [10, 100]$ cyc/mm, with $\langle v \rangle = 30$ m/s and a Gaussian scene spectrum $S(f_s) = \exp[-(f_s/50)^2]$ as a default, and returns a single scalar. Each tool exposes an MTF query at user-specified spatial frequency; the stubs differ only in syntax and integration with the host application’s optimization loop. The Python file `event_mtf_operand.py` in the open-source repository (Appendix B) provides the canonical reference implementation that all four stubs reduce to.

(a) CODE V Macro-PLUS. The radial MTF at frequency $\hat{f}$ is queried via the built-in `MTFV` function in CODE V’s Macro-PLUS language [18]. A custom merit-function operand is registered by calling the macro from a user-defined `SCP` (script) operand:

```
! event_mtf.seq — Event-MTF operand for CODE V
NUM ^v_mean = 30.0      ! mean edge velocity, m/s
NUM ^f_lo   = 10.0      ! lower spatial freq, cyc/mm
NUM ^f_hi   = 100.0     ! upper spatial freq, cyc/mm
NUM ^n_pts  = 20
```

```

NUM ^df = (^f_hi - ^f_lo) / (^n_pts - 1)
NUM ^Fe = 0.0
FOR ^i 1 ^n_pts
NUM ^f      = ^f_lo + (^i - 1) * ^df
NUM ^mtf_v  = MIFV f1 g1 ^f r 0 0      ! radial MTF at ^f
NUM ^S_f    = EXP(-(^f / 50.0) ^^ 2)    ! scene spectrum
^Fe = ^Fe + ^mtf_v * ^f * ^v_mean * ^S_f * ^df
END FOR
WRI "Event-MTF Fe = " ^Fe

```

(b) Zemax OpticStudio via ZOSPy (Python). ZOSPy bridges Python to OpticStudio's COM interface; the MTFa operand returns sagittal/tangential MTF at user-specified frequencies. The integration loop is in pure Python, which is convenient for evaluating $p(v)$ -ensembles and $S(f_s)$ variants without recompiling the merit function:

```

# event_mtf_zospy.py — Event-MTF via ZOSPy
import numpy as np
import zospy as zp

zos = zp.ZOS()
oss = zos.connect_as_extension()
TheSystem = oss.TheSystem

f_arr  = np.linspace(10.0, 100.0, 20)      # cyc/mm
v_mean = 30.0                             # m/s
S_f     = np.exp(-(f_arr / 50.0) ** 2)     # scene spectrum

mtf_vals = np.zeros_like(f_arr)
for i, f in enumerate(f_arr):
# MTFa: 0=sagittal, 1=tangential; use sagittal here
mtf_vals[i] = TheSystem.MFE.GetOperandValue(
zp.constants.Editors.MFE.MeritOperandType.MTFa,
1, 0, f, 0.0, 0.0, 0, 0
)

Fe = np.trapz(mtf_vals * f_arr * v_mean * S_f, f_arr)
print(f"Event-MTF_Fe={Fe:.6f}")

```

(c) LightTools (LSL). LightTools is the natural environment for the joint Event-MTF + PST merit, because LightTools' stray-light analysis returns PST directly [17]. The combined merit penalizes designs whose PST exceeds the event-optics threshold of 10^{-5} :

```

# event_mtf_pst.lsl — LightTools Scripting Language
set vMean = 30.0
set fLo   = 10.0
set fHi   = 100.0
set nPts  = 20
set df    = (fHi - fLo) / (nPts - 1)

```

```

# Run stray-light analysis to obtain PST
runStrayLightAnalysis()
get pstValue = StrayLight.PST

set Fe = 0
for i = 1 to nPts
set f      = fLo + (i - 1) * df
set mtfVal = computeRadialMTF(f)
set Sf      = exp( -(f / 50.0) ^ 2 )
set Fe      = Fe + mtfVal * f * vMean * Sf * df
endfor

# Combined merit: Event-MTF penalized by PST overshoot
set pstPenalty = max(0, pstValue * 1e5 - 1)
set merit      = Fe / (1 + pstPenalty)
print "Combined Event-MTF + PST merit = ", merit

```

(d) RSoft FullWAVE / BeamPROP. RSoft is the natural environment for the (Q-E) cell, where wave-optical PSF tails matter [13]. The flow runs FDTD or BPM, extracts the field at the detector plane, computes the radial MTF, and integrates as above:

```

# event_mtf_rsoft.py — Event-MTF from FDTD-derived PSF
import numpy as np
from rsoftpy import FullWAVE # vendor binding (placeholder name)

sim = FullWAVE("lens_model.ind")
sim.run()
psf = sim.get_intensity_at_detector() # 2D ndarray

# Optical transfer function via 2D FFT
otf_2d = np.fft.fftshift(np.fft.fft2(psf))
mtf_2d = np.abs(otf_2d) / np.abs(otf_2d).max()

# Radial average to get MTF(f)
h, w = mtf_2d.shape
yy, xx = np.indices((h, w))
r = np.sqrt((xx - w / 2) ** 2 + (yy - h / 2) ** 2).astype(int)
mtf_radial = (np.bincount(r.ravel(), mtf_2d.ravel())
/ np.bincount(r.ravel()))

# Map pixel-frequency index to cyc/mm (depends on sim grid)
pix_to_cycmm = 1.0 # set from sim.grid_spacing
f_index = np.arange(len(mtf_radial)) * pix_to_cycmm

f_arr = np.linspace(10.0, 100.0, 20)
v_mean = 30.0
S_f = np.exp(-(f_arr / 50.0) ** 2)

```

```

mtf_at_f = np.interp(f_arr, f_index, mtf_radial)

Fe = np.trapz(mtf_at_f * f_arr * v_mean * S_f, f_arr)
print(f"Event-MTF_{Fe:.6f}")

```

Common pattern. All four stubs share the same five-line algorithmic core — query MTF on a frequency grid, multiply by $f_s \langle v \rangle S(f_s)$, integrate, normalize, return scalar. The Maréchal cross-check (Appendix B) applies unchanged: at zero aberration, F_e should saturate to its diffraction-limited reference value, and at small σ_w the ratio F_e/S^β should fall in $[0.5, 0.7]^{-1}$ as predicted by Section 5.2 [16]. Designers using these stubs should validate against the Maréchal cross-check before incorporating F_e into a production merit function.

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